

Chapter - Documentation

Software Documentation

Definition:

Software Documentation is written text or illustration that explains how a software operates, how it can be used, and how it should be maintained. It provides necessary information for different people involved with the software, such as developers, testers, and end users.

Types of Software Documentation:

Software documentation is mainly divided into:

1. Requirement Specification
2. Architecture / Design Documentation
3. Technical Documentation
4. User-End Documentation
5. Marketing Documentation

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1. Requirement Specification

Requirement Specification describes the requirements, expectations, and goals of a software system.

Types of Requirement Specification:

1. Business Requirement Specification (BRS)
2. Functional Requirement Specification (FRS)
3. Software Requirement Specification (SRS)

1.1 Business Requirement Specification (BRS)

Definition:

Business Requirement Specification (BRS) is a formal document that describes the `business needs and objectives provided by the customer or stakeholders.` It is created at the beginning stage of software development and focuses on what the client wants to achieve from the software. It provides a high-level view of all business requirements.

Example:

A company wants to build an online banking system. BRS will describe goals like online money transfer, reducing manual banking work, and improving customer service.

1.2 Functional Requirement Specification (FRS)

Definition:

Functional Requirement Specification (FRS) is a document that ***describes all the functions and operations that a software system must perform.*** It explains the step-by-step activities of the system and defines how different software components should behave during user interaction.

Example:

For a university management system, FRS describes functions like student login, course registration, result checking, and assignment submission.

1.3 Software Requirement Specification (SRS)

Definition:

Software Requirement Specification (SRS) is a document prepared by ***system analysts*** that describes the complete requirements of a software system. It contains both functional and non-functional requirements and works as an

agreement between stakeholders and developers. ***It provides a framework that guides the entire software development process.***

Example:

For an e-commerce website, SRS includes requirements like user registration, product search, online payment, security, and performance requirements.

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2. Architecture Documentation

Definition:

Architecture Documentation provides a ***complete overview of the structure and design*** of a software system. It represents different architectural views and explains how different components of the system are organized and connected.

Includes:

- ERD (Entity Relationship Diagram)
- UML Diagram
- DFD (Data Flow Diagram)

Example:

In a hospital management system, architecture documentation shows database relationships using ERD and system interactions using UML diagrams.

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3. Technical Documentation

Definition:

Technical Documentation describes the ***technical aspects*** of a software project. It contains detailed information that helps developers build, modify, and maintain the software. It may include information about ***libraries, dependencies, APIs, and source code.***

Example:

A developer guide containing ***API documentation, database details,*** and ***programming instructions*** is technical documentation.

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4. User-End Documentation

Definition:

User-End Documentation is documentation ***prepared for end users*** to help them understand and use a software product or service. It provides simple instructions and guidance about software features and operations.

Includes:

- Snapshots
- Diagrams
- Tutorials
- Steps
- Do’s and Don’ts

Example:

A mobile app user manual explaining how to create an account, change settings, and use features.

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5. Marketing Documentation

Definition:

Marketing Documentation refers to ***promotional materials*** that are created to attract potential customers and encourage them to use a product or service. ***It describes product features, benefits, and advantages compared to alternatives.***

Example:

A software company's website, brochure, or advertisement describing the features of their application.